Matrices, Transformations & Eigenstructure

Mathematics Foundations — Session 2

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Today's Goal

- See a matrix not just as a grid of numbers, but as something that acts on space
- Multiply matrices by hand and trace a vector through the forward pass of a small network
- Read eigenvectors as the directions a transformation leaves pointing the same way, with PCA as the ML use case

→ Deliverable: a by-hand two-layer forward pass, plus naming the max-variance eigen-direction on a small dataset

Plan

1. Opening Quiz
2. Matrix Definitions & Base Operations
3. Linear Transformations
4. Eigenvalues & Eigenvectors
5. Machine Learning
6. Recap
7. To Do for Next Session

Opening Quiz

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Opening Quiz: Vectors & the Dot Product

Matrix Definitions & Base Operations

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What Are Matrices?

Definition

Matrices: Example and Elements

Matrices in Machine Learning

Matrix Properties

Matrix–Vector and Matrix–Matrix Multiplication

Is $AB = BA$?

Example

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$

Compute AB and BA.

Multiplying Matrices of Different Sizes

Example

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 3 & 2 & 0 \end{bmatrix}$$

Compute AB.

Special Matrices

Identity Matrix

Transpose

Zero Matrix

Symmetric Matrices

Orthogonal Matrices

Vocabulary

Exercises (1/2)

Given matrices

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

1. Calculate $A + B$
2. Calculate $A - B$
3. Calculate AB

Exercises (2/2)

Given matrix

$$A = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$$

1. Calculate the transpose A^T
2. Calculate $A + A^T$
3. Is A symmetric? Justify.

Linear Transformations

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Introduction

What Is a Linear Transformation?

Definition

A linear transformation is a function that preserves vector addition and scalar multiplication:

$$f(a+b) = f(a) + f(b), \qquad f(ka) = kf(a)$$

Linear Transformations and Geometry

The Determinant

Exercise: Determinants

Problem. Given matrices:

$$A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}$$

1. Calculate $\det(A)$ and $\det(B)$
2. Calculate $\det(AB)$ and verify that $\det(AB) = \det(A) \cdot \det(B)$
3. Determine if matrices A and B are invertible
4. Calculate $\det(A^T)$ and verify that $\det(A^T) = \det(A)$

Rank

Eigenvalues & Eigenvectors

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Eigenvalues

Eigenvectors

Definition

An eigenvector of a matrix A is a non-zero vector v such that:

$$Av = \lambda v$$

where λ is called the eigenvalue.

Exercise: Eigenvalues and Eigenvectors

Problem. Find the eigenvalues and eigenvectors of:

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

Machine Learning

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The Forward Pass

PCA

Deliverable

A by-hand two-layer forward pass, plus naming the eigen-direction of maximum variance on a small dataset.

Recap

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Recap

To Do for Next Session

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To Do for Next Session